

Parenthood and the Gender Gap in Pay

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We compare the income and wage trajectories of women to those of their male partners before and after parenthood. Focusing on the within-couple gap allows us to control for both observed and unobserved attributes of the spouse and to estimate both short- and long-term effects of entering parenthood. We find that 15 years after the first child has been born, the male-female gender gaps in income and wages have increased by 32 and 10 percentage points, respectively. In line with a collective labor supply model, the magnitude of these effects depends on counterfactual relative incomes or wages within the family.

I. Introduction

The Nordic countries are characterized by high female labor force participation, a generous social insurance system, and a glass ceiling: females

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do not reach the best-paid positions. The glass ceiling is claimed to be related to the generous parental leave system, in which the replacement rate is proportional to lost earnings. The reasoning is that women with young children are encouraged to participate in the labor force, but to do so less intensively than comparable men, since women are more involved in child rearing. In consequence, women may choose less demanding jobs, leading to a glass ceiling (Albrecht, Björklund, and Vroman 2003)

In the present paper, we pick up on the importance of parenthood addressed in the previous literature. Arguably, career interruptions due to parental leave, or perhaps more importantly, greater continuing child-rearing responsibilities, have an impact not only at the top but also across the whole earnings distribution. The aim of this paper is therefore to estimate the short- and long-term effects of entering parenthood on the gender gaps in income and wages in Sweden. Our focus is on women's income and wage trajectories in relation to their partners' before and after the birth of the first child. In contrast to earlier studies estimating the effect of parenthood, we use neither variation across women or men with and without children nor variation in the timing of when a particular female or male enters parenthood (see, e.g., Waldfogel 1997; Budig and England 2001; Sasser 2005; Kennerberg 2007; Loughren and Zissimopolus 2008). Instead, we explore the within-family variation over time. The basic identification assumption is that the decision of when to enter parenthood is not induced by unobservable information of a changed direction of the income and wage trajectory of one of the spouses.

The empirical analysis is based on universal administrative Swedish registers. These data allow us to match mothers and fathers with each other, and our unit of observation is parent couples. We are able to track parents' incomes and wages over a significant part of their labor market career, starting a few years before parenthood until about 15 years after the birth of the first child. The reported wage is for a full month, assuming full-time, even if a person works part-time, while the labor income measure reflects both wages and actual hours worked.

An obvious advantage with our approach is that we can draw direct inference on the average gender gap rather than on female earnings or wages. In this way, we are able to control for observed as well as unobserved attributes of the spouse, which are ignored in most previous studies aiming at estimating the effect of parenthood. The attributes of the spouse, however, have been shown to be important for the labor supply after parenthood (Bertrand, Goldin, and Katz 2010; Bertrand, Kamenica, and Pan 2013).

Another advantage is that we can directly study the importance of comparative advantages within the household. Theoretically, we think of the household's underlying labor-supply decision process in terms of a collective labor supply model, where the bargaining power of each household member is partly determined by comparative advantages in the labor mar-

ket (Chiappori 1992). To see whether the data support this model, we study whether the effect of parenthood on labor supply varies with each household member's income or wage potential in the absence of a child, and if so, by how much. To this end, we use quantile regressions to study how any potential effect of parenthood differs across the distribution of the gender gap in income and wages in the absence of having a child.¹

Our main finding is that 15 years after the first child has been born, the percentage difference between men's and women's incomes increases by 32 percentage points from the pre-child difference, while the corresponding effect for monthly wages is 10 percentage points. Using quantile regressions, we find that the effect of parenthood on the male-female gap increases with the counterfactual gap in incomes and wages the couple would have experienced in the absence of parenthood. For the large majority of couples, parenthood implies an increased gender gap in incomes and wages, but for the 20% of the couples with the smallest or even a negative counterfactual male-female gap, the effect is negative. Moreover, we estimate the effect of parenthood separately for couples with varying relative pre-child education. Consistent with theory, the effect on the gender gap is higher the less educated a female is relative to her spouse. The general message of this analysis is that the match type is crucial for the magnitude of the effect of parenthood on incomes and wages: comparative advantages in terms of earnings potential are important for how the monetary costs of parenthood are shared between the parents.

The estimated effect of parenthood can be attributed either to mothers' career interruptions in direct proximity to childbirth (i.e., parental leave) or to their long-term continuing responsibilities for child rearing (e.g., by working part-time). Although we cannot directly determine which mechanism is causing the long-term effect, our interpretation of the separate results on income and wages is that the latter mechanism is the main effect source. For income, we observe a shift in the gender gap due to females' decrease in hours worked directly after childbirth. For wages, however, there is no shift but a gradual relative increase in the gender gap, which can likely be attributed to mothers' continuing responsibilities for child rearing.

¹ Within-household specialization is discussed in, e.g., Manser and Brown (1980), McElroy and Horney (1981), and Konrad and Lommerud (1995). We do not claim that the gender division of household work is solely determined by comparative advantages in the labor market. Recent studies by Tichenor (1999), Booth and van Ours (2005), Evertsson and Neramo (2007), and Boye (2008) have presented evidence that women do relatively more unpaid household work than men also when the incomes are more equally distributed within the household. Studies in gender theory explain this by the importance of gender norms (see, e.g., West and Fenstermaker 1995). Akerlof and Kranton (2000) instead stress the importance of gender identity. Nevertheless, our results clearly indicate that comparative advantages in the labor market play a very important role (see Sec. VI).

The rest of this paper has the following structure. In the next section, we provide information on the evolution of the female labor supply and discuss some special features of the Swedish labor market. Section III contains the identification strategy. Section IV presents the data, descriptive statistics, and some graphical analysis. The baseline results are presented in Section V, and in Section VI, we present results on the importance of comparative advantages within the household. Section VII contains a discussion of the results in relation to macrodata evidence and to the previous literature, and, finally, Section VIII concludes the paper.

II. The Gender Gap and Family Responsibilities

This section contains a brief discussion about the gender gap in income and wages in Sweden during the past decades from an international perspective. Figure 1 shows the female labor force participation in eight European countries for the period 1983–2008. We can see that the participation rate in Sweden, as in the other Nordic countries, has been high and rather constant during the past decades.²

Several reforms since the beginning of the 1970s have contributed to the high labor supply of women. The introduction of the individual tax system in 1971, whereby taxation of spouses was individualized, created large incentives for Swedish women to participate in the labor force. Selin (2014) concludes that the female labor supply increased by 10 percentage points due to this reform.³

The parental leave system was introduced around the same time, in 1974. The replacement rate for parental leave was from the very beginning proportional to forgone earnings, which probably has contributed to the high employment rate among women before entering parenthood. The generous replacement rate for parental leave and the flexibility of when to use the paid days probably have also contributed to the fact that most Swedish mothers in the labor market work while having small children.⁴ During the child's first 18 months, any one of the parents can stay at home on a full-time basis with paid parental leave and job protection. Parents can take turns being on parental leave, as long as the total number of months on leave is at most 18 months per child. Thereafter, parents are allowed to reduce their working hours up to 25% until the child turns 8 years old

² In the early 1990s, Sweden experienced an economic crisis, which drastically decreased the labor supply of both men and women. Public expenditures were cut dramatically. Since women are overrepresented in the public sector, this crisis affected women to a somewhat larger degree than the males.

³ In this context it is interesting to note that, among the compared countries, Germany has the lowest female participation rate and in Germany couples are still taxed together.

⁴ The introduction of the Swedish parental leave system has not (to our knowledge) been evaluated with respect to its effect on the female labor supply.

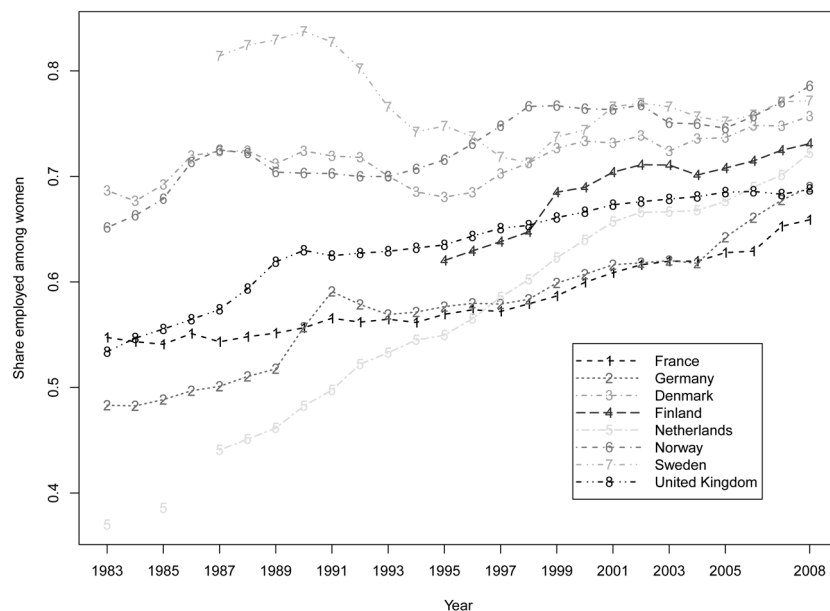


FIG. 1.—Labor force participation rate among women in eight European countries between 1983 and 2008. The numbers are from the EU Labor Force Survey conducted by Eurostat. The employment rate is calculated by dividing the number of persons aged 20–64 in employment by the total population in the same age group. The survey covers the entire population living in private households and excludes those in collective households such as boarding houses, halls of residence, and hospitals. The population of employed consists of persons who during the reference week did any work for pay or profit for at least 1 hour or were not working but had jobs from which they were temporarily absent, for instance, due to parental leave.

(SFS: 1995:584. Föräldraledighetslag [Swedish law 1995:584. Parental leave law]). Women use the parental insurance most: they take 80% of the paid parental leave days (Försäkringskassan 2011). In addition, 44% of all women aged 25–54 work part-time (less than 35 hours per week). The corresponding share of men who work part-time is 10%.⁵

In parallel to the institutional changes described above, there has been a rapid increase in the public provision of child care, especially during the 1980s. This may also have contributed to the high female labor supply, but it could also be a symptom of an increased demand for child care; a causal relation between public provision of child care and female labor participation has not yet been established.

⁵ The source is public statistics from Statistics Sweden, published on the Web: http://www.scb.se/Pages/Article____332715.aspx.

Unfortunately, there are no public statistics on how the age composition of the labor force changed over time before 1980. However, our own calculations from publicly published data from Statistics Sweden show that in the beginning of the 1980s, the increase in female labor force participation was mainly driven by women with pre-school children. From 1990 onward, the participation rate among women with small children (ages less than age 7) was even higher than among women in general (see fig. 2).⁶ However, women with small children still spend more effort and time at home than do men with small children. This unequal gender division at home most likely has consequences for the remaining effort in the labor market.

III. Identification and Empirical Strategy

The choice of forming a family, which we define as having a child together, is most often planned, both as to with whom and as to its timing. We think of a matching market where the matching depends on the partner's current and expected future productivity. Women carry the child for almost a year and also take on the main burden of the rearing of the child. It is therefore reasonable to think of the decision of having a child as depending on the insurance coverage for the income loss during pregnancy and the child's early years.⁷ Furthermore, the timing of when to have a child is probably a function of the degree of labor market attachment. For instance, with income replacement proportional to the income loss while on parental leave (as is the case in Sweden), it is important to be employed when entering parenthood. In the estimation of the effect of parenthood, it is therefore important to control for factors correlated with both future incomes and the determinants for the decision of becoming a parent.⁸

The empirical strategy presented below allows us to identify the effect of parenthood on the gender gap in incomes and wages under the following optimizing behavior of the agents within the family: (i) the decision of having

⁶ This is most likely a cohort effect; that is, almost all young women participate in the labor force, while being a housewife is more common among older women.

⁷ However, in the Swedish context, with quite high social insurance coverage and with parental benefits tied to pre-birth income, this kind of within-family insurance solution may be less dominating than in countries with less social insurance coverage.

⁸ An alternative is, of course, to find an instrument for parenthood. Seminal papers include Angrist and Evans (1998), which uses parental preferences for mixed sibling-sex composition as a source of exogenous variation in family size, and Bronars and Grogger (1994), who employ twin births. The mixed-sex strategy measures the effect of a third child (compared to having two children), while using twin births: the focus is on the effect of two children (compared to one). There have also been a few papers using instruments in the estimation of the effects for first-time mothers. Hotz, McElroy, and Sanders (2005) and Miller (2011) both use variation from miscarriages. Miller (2011) uses conception as well. Klepinger, Lundberg, and Plotnick (1999) exploit age at menarche and regional policy variation in welfare generosity and access to family planning services.

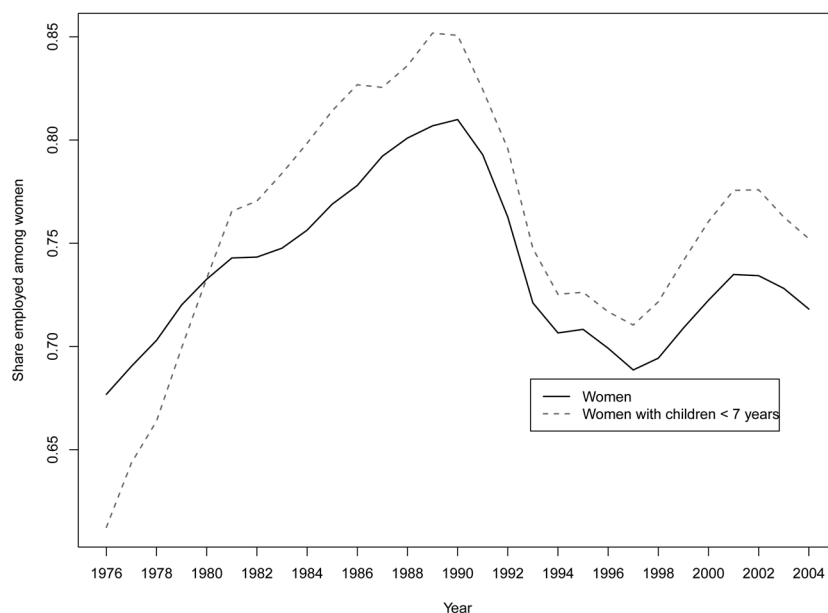


FIG. 2.—Labor force participation between 1974 and 2004 among Swedish women aged 16–64 with and without children less than 7 years old. The figures are based on the Swedish Labor Force Survey (AKU). AKU is conducted by Statistics Sweden among 29,500 randomly sampled residents in Sweden aged 20–64.

a child is based on expectations about the future income of the household, and (ii) parents solve their dual career problem by equalizing the marginal utility of time and effort in home production and market work (Becker 1985). In this setting, a couple will have a child if the utility of having the child is greater than the expected loss of future household income.⁹

In the present paper, we condition on having a child, which means that the condition above is satisfied for the couples in our analysis. The question then is whether the expected income loss is unequally split within a couple. Under the assumption that the timing of parenthood is not based on expected changes in the within-couple income gap in the absence of a child, the effect of parenthood can be consistently estimated provided that we control for the pre-child income or wage difference (see app. A for details).

⁹ This is intuitive, but it can also be shown formally. Assume that utility is additively separable in consumption and in pure preferences for having a child and that households maximize their utility in two steps. First, utility is maximized conditional on (i) having a child or (ii) not having a child. In the second step, the household decides on whether to have a child, based on the indirect utility in either state. The result from this simple model is that a household will have a child if the utility of having the child is larger than the expected utility loss of future household income.

The identification strategy can be seen as a form of a difference-in-differences framework. Both groups (men and women) are affected by the treatment, but we allow the treatment magnitude to differ between the two groups. The identifying assumption is the same as in a traditional difference-in-differences setting; that is, the intervention (parenthood) must be strictly exogenous. Because we sample parents, strict exogeneity in our case is a statement about the timing of parenthood rather than parenthood per se.

Since we have data for several time periods before parenthood, we can also control for potential trends in our dependent variable. The main identification assumption is that the timing of when to have a child should not be determined by expected future shocks to the within-family gender difference in income or wage the couple would have experienced in the absence of entering parenthood. This means that the timing of parenthood should not be influenced by, for us, unobservable information about future income or wage trajectories of men in comparison with women, or vice versa. For example, if the timing of having a child is determined by a promise of a promotion or advancement in the career of men but not of women, then our identification strategy fails. If anything, since parental benefits are tied to forgone earnings, the timing of parenthood may depend on positive shocks to the woman's income. However, if these shocks take place before the birth of the child, they are observed in our data and hence controlled for.

IV. Data and Descriptive Statistics

We now present the data and some descriptive statistics including some graphical evidence of the effect.

A. Data

The data used in the analysis are taken from administrative registers covering all residents in Sweden between 16 and 65 years of age during the period from 1986 to 2008. From these registers, we have information about sex, age, labor market income, education, and family relations. We can link parents to their biological children, and we have access to birth year and birth order. Couples are traced from information about adults who are registered as having at least one child together.

Our two outcome variables are *yearly labor income* and *full-time equivalent monthly wages*. Information about labor market income is based on the annual reports from the employers to the tax authorities. Thus, this amount includes the total individual pre-tax income from work, and it does not include the amount of paid parental leave, tax reductions, or social transfers, such as means-tested cash benefits. The income measure does, however, include a part of the sick-pay: in case of illness, the first day is not replaced. Thereafter, the employer pays sick-pay for the 14 following

days, and this amount is included in the income measure. The replacement rate has varied over time between 75% and 90% of lost earnings, up to a limit that in 2009 corresponded to earnings in the 70th percentile of the earnings distribution. After 14 days, the Swedish Social Insurance Agency disburses sickness benefits, which are not included in the labor income measure.

The pre-tax monthly wages that we use have two main components: a fixed wage and variable pay, such as performance-based bonuses, night-shift and on-call allowances, and so forth. We observe the total wage and not the individual components. The reported wage is for a full month, assuming full-time, even if a person works part-time. Until the year 2000, Statistics Sweden defined full-time as 165 hours/month for all employees. For later years, there are minor differences in the definition of full-time across different sectors, depending on the relevant collective agreements between unions and employers.

The fact that the wages are reported assuming full-time implies that the wage variable does not reflect differences in working hours during the month. On the other hand, labor income does reflect working hours, and thus it serves as a combined measure of wage and labor supply. We do not have a direct measure of working hours in the data.

For the public sector, we have annual wage information for all who were employed on the first day of September, October, or November each year.¹⁰ For the private sector, we have information on wages from a yearly 50% random sample, measured some time between July and October each year.¹¹ Thus, while our income data as well as wages for publicly employed cover all individuals, wage data for the private sector consist of a cross-sectional representative sample each year.

From these data, we define couples who had their first-born child together during any of the years 1990 to 2002. We then track these couples' labor market activities before and after the birth of their first child. This implies that we can track each couple for at least 4 years before and 6 years after the birth of their first child. Parents who had their first child in 1990 are tracked at most 18 years after the birth of their first child, while parents whose first child was born in 2002 are tracked at most 16 years before the birth year of their first child. Table 1 summarizes the period of coverage of our data, measured in years.

¹⁰ The date of measurement depends on whether the employer is a municipality or county council (November 1), the state (September 1), or a university or college (October 1). A person who has an employment some time during the year but no employment on the relevant measurement date will have a missing observation for the wage variable during that specific year.

¹¹ There are 11 different groups of private employers, roughly corresponding to different industries. September and October are the most common measurement months, and July is an exception valid only for the construction sector. However, the measurement period for individuals employed in the construction sector includes not only July but also August and September.

Table 1
Data Coverage: Number of Years That We Can Follow Parents

	Year of Birth		
	1990	1997	2002
Before birth year	4	11	16
After birth year	18	11	6

NOTE.—In the analysis, we use data for parents giving first birth in 1990–2002. Intermediate years are not shown here in order to save space.

The panel enables us to study the dynamics of the within-couple difference in incomes and wages, that is, the variation over the years before and after the birth of the first child. Thus, we are able to draw conclusions about both the short- and long-term effects of entering parenthood.

B. Descriptive Statistics

In the analysis, we restrict the population to couples where both men and women have positive income 2 years before their first child is born.¹² Table 2 presents the descriptive statistics (means and standard deviations) for this population for the variables used in the analysis, both for levels and within-couple differences.

From the table, we can see that the mean age when entering parenthood has increased over the study period. Among mothers, the average age when entering parenthood was about 24 years in 1990 and it was 2–3 years higher in 2000, depending on completed fertility. There is a similar increase among men. However, the male-female age difference of approximately 2 years is stable.

The number of years of education ranges between 11 and 12.5 years. Parents who have more than one child are on average more educated than those with just one child, and mothers are in general more educated than fathers. The education level as well as the gender difference are both increasing over time.

The mean pre-birth income and wages are higher for men than for women. Parents who have more than one child have higher incomes and wages than those with just one child. Income and wages are increasing over time, and the gender difference is increasing in absolute value over time.¹³ The higher average income and wages of men are accompanied with larger standard deviations. In other words, the variation in the income and wages among men is greater than that for women.

¹² The analysis was repeated with a more restricted sample, consisting of fathers and mothers with an annual income of more than 50,000 SEK (about \$6,329) 2 years before the first child was born. The results for this sample were qualitatively the same as the results displayed below.

¹³ Incomes and wages are measured in 2008 prices, so these changes reflect increases in real income.

Table 2
Data Description of All Variables Measured 2 Years before Birth Year

	Birth Year, First Child Completed Fertility					
	1990		1995		2000	
	One Child	≥ Two Children	One Child	≥ Two Children	One Child	≥ Two Children
A. Full Sample on Income						
Age, father	26.5 (5.97)	26.2 (4.42)	28.7 (6.09)	27.5 (4.38)	29.2 (6.17)	28.3 (4.33)
Age, mother	24.3 (5.55)	24.0 (3.85)	26.6 (5.56)	25.4 (3.84)	27.1 (5.68)	26.3 (3.86)
Age, (father – mother)	2.2 (3.98)	2.21 (3.35)	2.09 (4.33)	2.12 (3.38)	2.11 (4.3)	2.02 (3.41)
Education, father	11.1 (1.97)	11.7 (2.07)	11.7 (2.12)	12.2 (2.1)	11.9 (2.1)	12.5 (2.15)
Education, mother	11.0 (1.91)	11.7 (1.93)	11.7 (1.95)	12.3 (1.94)	12.0 (2.18)	12.9 (2.13)
Education (father – mother)	.0977 (2.15)	–.0427 (2.15)	–.0701 (2.16)	–.112 (2.08)	–.0954 (2.28)	–.39 (2.22)
Income, father (SEK)	175,276 (97,680)	195,255 (93,570)	182,222 (119,723)	193,130 (117,609)	218,025 (143,701)	242,405 (151,378)
Income, mother (SEK)	130,307 (75,672)	146,055 (68,341)	140,200 (88,790)	148,295 (79,566)	155,407 (108,939)	177,646 (105,680)
Income (father – mother; SEK)	44,969 (95,113)	49,199 (96,575)	42,022 (119,619)	44,835 (120,605)	62,618 (138,953)	64,760 (153,899)
No. of couples	6,755	29,185	4,961	19,955	4,981	20,522
B. Restricted Sample on Wages						
Wage, father (SEK)	18,546 (5,646)	18,707 (5,224)	19,165 (5,876)	18,785 (5,023)	22,582 (6,918)	22,918 (7,410)
Wage, mother (SEK)	15,423 (3,266)	15,311 (3,127)	16,347 (3,725)	16,041 (3,007)	18,899 (4,801)	18,994 (4,767)
Wage (father – mother; SEK)	3,044 (5,693)	2,994 (5,169)	2,688 (5,948)	2,459 (4,836)	3,330 (6,707)	3,501 (7,028)
No. of couples	857	4,706	940	4,248	1,001	4,776

NOTE.—Yearly labor income and full-time equivalent monthly wages are measured in SEK in 2008 prices. Full-time is approximately 165 hours/month, and the December 2008 exchange rate was approximately 7.9 SEK/USD. Standard deviations are in parentheses.

Finally, the standard deviations of the within-couple differences in table 2 are smaller than the standard deviations of the father. This fact provides evidence of a positive assortative mating in Sweden. The presence of positive assortative mating implies that it is important to control for unobservables that are potentially correlated with both parenthood and the dependent variables of interest.¹⁴

C. Graphical Evidence

Before presenting our formal analysis, we provide some initial evidence by graphically examining the raw within-couple gap in income and wages over time. The patterns shown in figures 3 and 4 reveal that before the birth of the first child, the income and wage trajectory of the parents are parallel, suggesting no gender-specific trend over time. Men have higher pre-birth wages and incomes. After the birth of the first child, there is a dramatic change. First, we have a sharp drop in the income of women due to their longer parental leave. After about 6 years, women are back in a track parallel to that of men but at a substantially lower level.

There is no corresponding drop in wages, suggesting that there is no direct relative wage penalty of being on parental leave.¹⁵ This means that women do not suffer a high immediate loss in human capital relative to their spouse due to short-term labor market interruptions. However, in the long run, the wage trajectory of mothers is much worse than that of men. In order to study the dynamics in more detail, we now turn to the formal analysis.

V. Results

In this section, we present the estimation results, starting with the average effect of parenthood for our sample at different time periods after the

¹⁴ Let y_m be the variable of interest for males and y_f be the corresponding variable for females. We have

$$\text{Var}(y_m - y_f) = \text{Var}(y_m) + \text{Var}(y_f) - 2\rho_{mf}\sqrt{\text{Var}(y_m)\text{Var}(y_f)},$$

where ρ_{mf} is the correlation coefficient between y_m and y_f . The sign of ρ_{mf} decides the sign of the last term. In particular, when we have positive assortative mating and $\rho_{mf} > 0$, we get $\text{Var}(y_m - y_f) < \text{Var}(y_m) + \text{Var}(y_f)$, as in table 2. For instance, we calculate a correlation coefficient of 0.46 between mothers' and fathers' pre-child wages. The degree of assortative mating in Sweden is in line with existing numbers for the United States; see figure 1 in Greenwood et al. (2014), presenting a value of 0.38 for the year 2002.

¹⁵ According to Swedish law, parental leave should not per se affect the wage negatively.

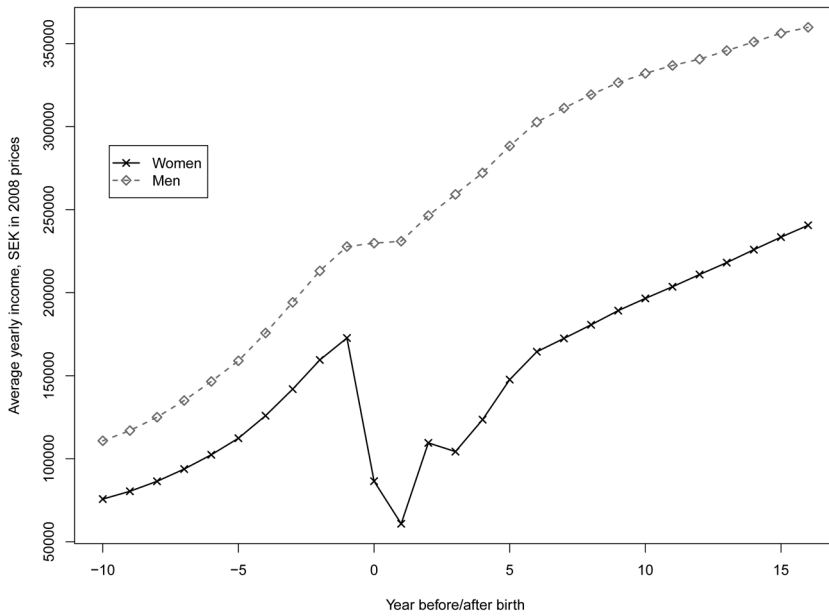


FIG. 3.—Average yearly income for women and men within matched couples before and after having their first child.

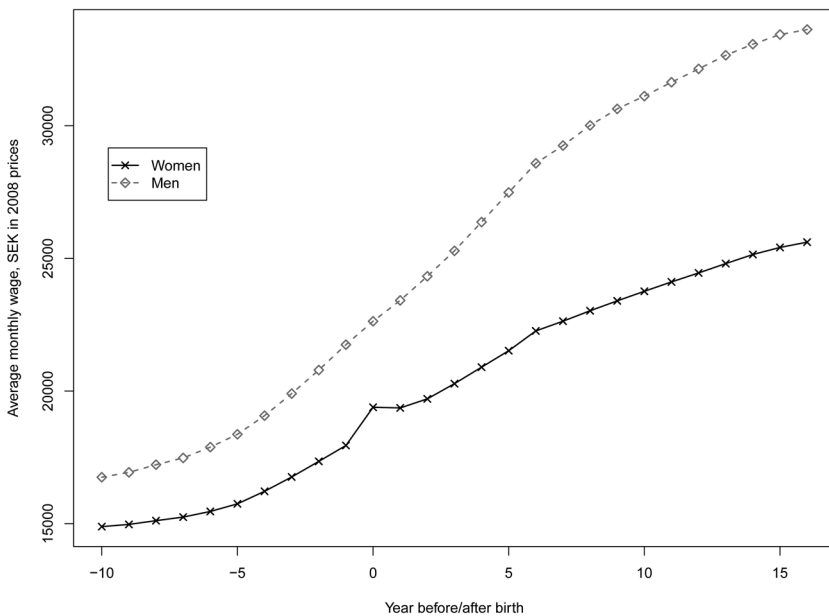


FIG. 4.—Average monthly wage for women and men within matched couples before and after having their first child.

birth of the first child. Then we turn to the heterogeneous effects estimated on different subgroups, as well as use quantile regressions for the whole sample. As explained in Section III, the effect of parenthood can be consistently estimated provided that we control for the pre-child income or wage gap. As in the traditional difference-in-differences setting, we could informally test for the strict exogeneity assumption by studying pre-parenthood trends. The result from this exercise shows that the maintained assumption seems plausible (see figs. 3 and 4).

To estimate the parameters of interest, we use the following specification:

$$\hat{y}_{tci} = \alpha + \sum_{j=0}^{18} \alpha_j \mathbf{1}[t = j] + \sum_{k=1990}^{2008} \psi_k \mathbf{1}[c = k] + \theta_0 \hat{y}_{(-2)ci} + \mathbf{x}'_{(-2)i} \boldsymbol{\beta} + u_{tci}, \quad (1)$$

where $t = -1, 0, \dots, 18$ denotes the year relative to child birth; the calendar year is indexed by $c = 1989, \dots, 2008$; $\hat{y}_{tci}$ is the within-couple log difference in income or wage t years after or before birth, observed in calendar year c ; i is a couple index; $\hat{y}_{(-2)ci}$ is the within-couple log difference in income or wages 2 years prior to child birth; $\mathbf{x}_{(-2)i}$ contains within-couple differences in age and pre-child years of education for couple i 2 years prior to child birth; $\mathbf{1}[\cdot] = 1$ if the expression within parentheses is true and 0 otherwise; and u_{tci} is an error term (for details on the identification, see app. A). Note that since we have variation in calendar times of child births, we are able to control for secular trends or shocks, ψ_k , in wage and income differences.

The parameters of interest α_j for $j = 0, 1, \dots, 18$ identify the effects of parenthood on the gender income gap at the year of birth and up to 18 years after, relative to the pre-child gender gap. By using the log transformation, we are essentially modeling the gender gap percentage. Thus the parameters identify, for each year after the birth year, the approximate change in the gender gap percentage compared to the pre-birth gap.¹⁶ The OLS estimator is consistent for α_j as long as u_{tci} is strictly exogenous. That is, conditional on any secular trend, within-couple pre-child gap, and pre-child covariates, the timing of childbirth cannot be based on expectations about future shocks to the gender gap trajectory in the absence of a child.

¹⁶ The log difference, which we use in estimation, is a good approximation of the percentage difference only for small differences. When discussing the results in the text, we therefore transform the parameters from log-points to percentage points by exponentiation. The formula is given by

$$pp_j = 100 \times (e^{\alpha_j} - 1) \quad \text{for } j = 1, 2, \dots, 18,$$

where α_j is from eq. (1).

The estimation results for income are presented in the left-hand part of table 3.¹⁷ We show the results with a varying conditioning set. It is quite obvious that we should control for the calendar year in the estimation. As our data cover a long period, we would like to control for different shocks in the economy that might affect women and men differently. However, from a theoretical point of view, it is difficult to argue that we should add couple-specific control variables. What matters is the information set of the couple at the time of decision. If the couple makes predictions that the wage profile is steeper for younger individuals (the women) or for more educated individuals, then we should control for these differences. It is, however, not obvious that the couples make these conditional predictions. We present the estimates from all three specifications: (i) without controls, (ii) with calendar-year effects, and (iii) with the full set of controls (see table 3). The results are reassuringly robust, as the parameters of interest in all three model specifications tell the same story and all estimates are statistically significant at the 0.10% level. Generally, including calendar year controls leads to higher effect estimates. The parameter estimates with the complete set of controls are somewhat lower. Since there is no qualitative difference in the results, we only discuss the results in the third and sixth columns, that is, the estimates with the full set of control variables.

We start with the labor income estimates, presented in the left-hand part of table 3. During the birth year (year 0), the parameter estimate of the change in the income gap is 1.35. The interpretation is that the gender gap in income is 135 log points (284 percentage points) larger during the first year of parenthood, in comparison to its pre-birth level.

What does this mean in absolute terms? Assume that in the absence of a child, the fathers' average income would have remained at the same level as 2 years before the birth. For the sake of an example, consider unconditional means instead of the conditional means implied by the regression. Using numbers from the description in table 2, the fathers and mothers of more than two children with their first child born in 1990 earned 195,255 SEK and 146,055 SEK, respectively (approximately \$24,716 and \$18,488, respectively). This implies a gender gap of about 34%. The parameter estimate of 135 log points (284 percentage points) would mean that the women's average yearly income shrinks to 46,747 SEK (\$5,917) during the birth year, that is, a gender income gap of 148,508 SEK (\$18,799).

During the year the child turns 1 year old, the parameter estimate is even higher (237 log points). An effect of this magnitude reflects the fact

¹⁷ The estimation was performed using ordinary least squares (OLS). The standard errors were estimated with a robust covariance matrix (White 1980). We have also estimated standard errors by clustering on household. The inference is qualitatively the same with the two types of standard error estimators.

Table 3
Yearly Effects of the Birth of the First Child in Year $t = 0$ on the Within-Couple Change in the Gender Gap in Income and Wages

	Yearly Income			Monthly Wages		
	No Controls	Calendar Year	Full Set of Controls	No Controls	Calendar Year	Full Set of Controls
Intercept	.0105 (.00401)	.0222 (.00909)	.0967*** (.00928)	.0506*** (.000726)	.0639*** (.00241)	.0798*** (.00254)
Year 0	1.339*** (.00703)	1.350*** (.00724)	1.345*** (.00724)	.0116*** (.00156)	.0126*** (.00158)	.00753*** (.00161)
Year 1	2.370*** (.00829)	2.376*** (.00861)	2.366*** (.00860)	.0204*** (.00138)	.0220*** (.00143)	.0203*** (.00144)
Year 2	1.425*** (.00819)	1.423*** (.00867)	1.408*** (.00866)	.0311*** (.00138)	.0342*** (.00146)	.0310*** (.00146)
Year 3	1.591*** (.00839)	1.584*** (.00899)	1.563*** (.00897)	.0401*** (.00150)	.0440*** (.00159)	.0392*** (.00158)
Year 4	1.221*** (.00817)	1.219*** (.00889)	1.193*** (.00888)	.0489*** (.00147)	.0546*** (.00159)	.0487*** (.00158)
Year 5	.920*** (.00802)	.951*** (.00888)	.920*** (.00886)	.0545*** (.00150)	.0612*** (.00165)	.0545*** (.00163)
Year 6	.743*** (.00797)	.811*** (.00897)	.773*** (.00895)	.0605*** (.00155)	.0686*** (.00172)	.0607*** (.00170)
Year 7	.663*** (.00826)	.740*** (.00929)	.696*** (.00927)	.0676*** (.00168)	.0762*** (.00185)	.0672*** (.00182)
Year 8	.577*** (.00855)	.674*** (.00963)	.623*** (.00961)	.0759*** (.00184)	.0846*** (.00201)	.0744*** (.00197)
Year 9	.505*** (.00891)	.625*** (.0100)	.567*** (.0100)	.0816*** (.00198)	.0915*** (.00216)	.0800*** (.00211)
Year 10	.436*** (.00933)	.574*** (.0105)	.510*** (.0105)	.0846*** (.00215)	.0953*** (.00233)	.0826*** (.00228)
Year 11	.379*** (.00982)	.535*** (.0111)	.466*** (.0111)	.0875*** (.00229)	.100*** (.00249)	.0858*** (.00243)
Year 12	.334*** (.0104)	.503*** (.0119)	.428*** (.0118)	.0895*** (.00247)	.105*** (.00270)	.0890*** (.00263)
Year 13	.293*** (.0112)	.466*** (.0127)	.385*** (.0127)	.0905*** (.00269)	.108*** (.00292)	.0914*** (.00285)
Year 14	.238*** (.0120)	.418*** (.0136)	.333*** (.0136)	.0940*** (.00299)	.114*** (.00323)	.0956*** (.00314)
Year 15	.173*** (.0133)	.367*** (.0148)	.279*** (.0148)	.0942*** (.00349)	.115*** (.00373)	.0955*** (.00363)
Year 16	.116*** (.0150)	.337*** (.0166)	.245*** (.0166)	.0973*** (.00407)	.120*** (.00432)	.0998*** (.00419)
Year 17	.0635*** (.0182)	.315*** (.0198)	.217*** (.0197)	.0976*** (.00538)	.120*** (.00561)	.0995*** (.00544)
Year 18	-.00490 (.0257)	.269*** (.0275)	.168*** (.0273)	.115*** (.00781)	.142*** (.00807)	.120*** (.00780)
$\ln(y_{m(-2)}/y_{f(-2)})$	.629*** (.00200)	.630*** (.00200)	.601*** (.00202)	.698*** (.00207)	.698*** (.00207)	.677*** (.00213)
Calendar year	No	Yes	Yes	No	Yes	Yes
x_{-2}	No	No	Yes	No	No	Yes

Table 3 (Continued)

	Yearly Income			Monthly Wages		
	No Controls	Calendar Year	Full Set of Controls	No Controls	Calendar Year	Full Set of Controls
<i>N</i>	5,244,327	5,244,327	5,244,327	451,883	451,883	451,883
No. of couples	368,122	368,122	368,122	51,097	51,097	51,097
<i>R</i> ²	.051	.052	.060	.294	.295	.330

NOTE.—Estimation is by ordinary least squares. Outcome variables are the within-couple log differences in labor income and wages. The full set of controls consists of calendar year dummies, within-couple age gap, pre-child within-couple differences in education, and pre-child log-differences in income and wages. Standard errors (in parentheses) are estimated with a robust covariance matrix (White 1980).
*** *p* < .001.

that women take out the largest part of the parental leave, and due to the design of the parental leave insurance in Sweden, there is an almost complete withdrawal of women from the labor market. After the second year since birth, the negative effect of parenthood on the gender gap in income decreases for every subsequent year, as women gradually increase their work hours. However, women never catch up: 15 years after the first child was born, the gender gap in income is still 32.2 percentage points higher as a result of parenthood compared to the pre-child gap.

Turning to the corresponding effects on wages, shown in the right-hand part of table 3, we have the following results.¹⁸ During the child’s first 2 years, the effect on the wage gap is relatively small compared with the effect on the income. This is consistent with either of the following two mechanisms, or most probably, a combination. First, high-wage mothers return to work faster after giving birth than do mothers with lower wages. Second, there is no direct relative wage penalty in Sweden for being on parental leave. However, in the long run, almost all Swedish mothers return to the labor market, and from about 10 years after the birth year and onward, the wage gap (relative to the pre-child wage gap) remains at a constant level of about 9–10 percentage points in favor of men.

To sum up, the following picture evolves when we combine the income and wage estimates. Immediately after childbirth, women’s relative hours

¹⁸ As noted in Sec. IV.A, data coverage on wages is 50% for the private sector, while there is full coverage for the public sector. Because of our setup, however, the sample size for our wage estimates is less than what the data coverage for wages would suggest. First, we condition on having a wage record 2 years prior to child birth. Second, both spouses have to have wage records at each value of the outcome variable. About 26% of the Swedish labor force work in the public sector (OECD 2011). Thus, the baseline probability of observing $(\ln \text{wage}_{mt} - \ln \text{wage}_{ft})$ for a specific couple is roughly equal to $[0.26 + (0.5 \times 0.74)]^4 \approx 0.16$. In addition, a person might not be working when sampled in November for several other reasons, e.g., unemployment.

worked seem to drop sharply.¹⁹ Many women appear to leave the labor force completely during the child's first year. In the somewhat longer run, the low magnitude of the wage estimates combined with the high magnitude of the income estimates suggest that women work part-time to a large extent as an effect of parenthood. In addition, although there is no substantial immediate wage penalty for parenthood, women's part-time work seems to imply slower human capital accumulation relative to their spouse, as the long-term effect of parenthood on the wage gap is considerable.²⁰

VI. Comparative Advantages at Home versus in the Labor Market

In this section, we are interested in heterogeneous effects with respect to the within-household earnings capacity in the absence of children, that is, the counterfactual distribution of relative incomes and wages over time within the household. Such an analysis reveals the importance of comparative advantages within the household with respect to labor market work versus home production (child care and household work). Comparative advantages could be captured by either/both investment in education (education is in most cases ended before entering parenthood) or/and earnings capacity (both instantaneous and prospective). Thus, we perform this heterogeneity analysis using two different, albeit related, approaches. First, we provide separate estimates for couples depending on the within-couple pre-child education gap. Second, we estimate quantile regressions, using the same specification (with the full set of controls) as in table 3.²¹

A. Matching with Respect to Pre-parenthood Education

In this analysis, we use data on completed education for the mother and father 2 years before the birth of the first child. The combinations of the mother's and father's theoretical years of education define our groups of interest.²² In order to have a sufficient number of observations in each education group, we restrict the analysis to 9 years of education (compulsory school), 12 years (upper secondary school), and 16 years (approximately a master's degree).

¹⁹ We do not have data on hours worked.

²⁰ An alternative mechanism is that part-time work is less valuable to the employer because of various fixed costs, difficulties in coordinating schedules, etc. However, if that were the case, we would expect to see a higher effect on wages during the first years after having a child, that is, when the extent of part-time work is at its peak. Instead, we see very small wage effects during the first years and an increasing effect in the longer run.

²¹ We obtain qualitatively very similar results without control variables.

²² The classification we use is the standard Swedish "Svensk Utbildningsnomenklatur (SUN)," which roughly follows the international ISCED 97 standard.

Consider a less educated female matched to a highly educated male. From a theoretical viewpoint, we would expect optimization on behalf of the couple to result in the female's taking greater responsibility for child rearing, relative to the average couple. Conversely, for a couple with a highly educated female and a less educated male, we expect the opposite. These predictions are confirmed by the data. The results for income are presented in figure 5. Because we present several estimates for each group, it is easiest to plot the results. To indicate significance at the 5% level, we superimpose each group number (1–7) on the curve whenever the particular estimate is statistically significant at the 5% level. For comparison, we also include the average effect estimates presented in the previous section.

The estimates show clearly how the effects vary with relative within-couple education. The effects for groups 2 and 3 (both consisting of highly educated males and less educated females) are both above the average effects curve, which is consistent with the theoretical prediction outlined

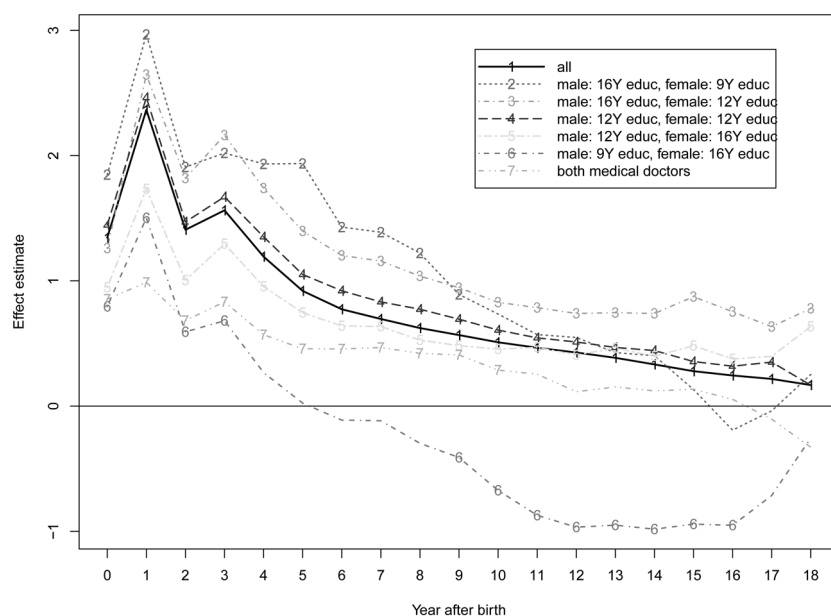


FIG. 5.—The effect of family formation on the change in the within-couple income gap: effect size by relative within-couple education and separately for medical doctors. Years of education and completed MD degree are measured at pre-child levels at $t = -2$. Estimates from OLS regressions with $\ln y_{imt} - \ln y_{ift}$ for $t \geq -1$ as outcome variable, where y_{imt}/y_{ift} is the yearly income of the male/female within each couple in year t . A parameter estimate is significant at the 5% level if the corresponding group number is superimposed on the curve. For instance, the parameter estimate for group 6 is significant for year 9 after childbirth, but insignificant for year 8.

above. Furthermore, the estimates for group 6 (where the relative male-female years of education is the lowest) is well beyond all the other groups', with a large part of the estimates being negative. In other words, the gender gap decreases, in many cases substantially, for couples where the female has the highest education relative to the male. Group 4, where both spouses have upper secondary school diplomas, is slightly above the curve representing the whole population. This is qualitatively consistent with the fact that women are on average more educated than men (see table 2). Thus, for an equally educated group, we would expect the effect on the gender gap to be slightly higher than for the average couple. Finally, the effect curve for group 5, where females are slightly more educated than males, is below the average effect curve before year 11, then crosses it, and is slightly above thereafter.

Figure 6 presents similar results for wages. However, due to the fact that wage observations do not cover the entire population, we lose precision. Once more, the effects for groups 2 and 3 (highly educated males and less educated females) are both above the average effects curve, the estimates for group 2 being even more pronounced than for income. The point estimates for group 6 (less educated males and highly educated females) are all below the average effects curve, as expected; however, none of those estimates is significant at the 5% level.

Finally, using both the level and field of education, we perform a separate analysis for medical doctors (see Sasser 2005).²³ In this case, theory does not provide much guidance concerning the size of the effects relative to the mean effect that we estimate for the whole population. At the level of precision that we have chosen, the mothers and the fathers have the same education either 1 or 2 years prior to child birth. The results from figure 5 show a lower effect of parenthood on the gender gap in income for the group of medical doctors (group 7) than for the whole population, when significant. Once more, the results for wages are more imprecise due to worse data coverage (see fig. 6). The long-term point estimates (after year 9) for doctors are below the average, although most are insignificant. Before year 9, several of the point estimates are significant and slightly above average. On the whole, however, it is difficult to draw any conclusions from the results for doctors in figure 6.

B. Quantile Regressions

For each year after the birth of the first child, we estimate the effect of parenthood in the 10th to 90th percentile of the conditional distribution of the within-couple counterfactual gaps in the income and wage trajectories.

²³ Before 2000, the SUN education codes for medical doctors were 66000–66199, excluding 66002. Starting from 2000, the education standard was changed, and the new codes are 557 for education level and 721a–721k for education field.

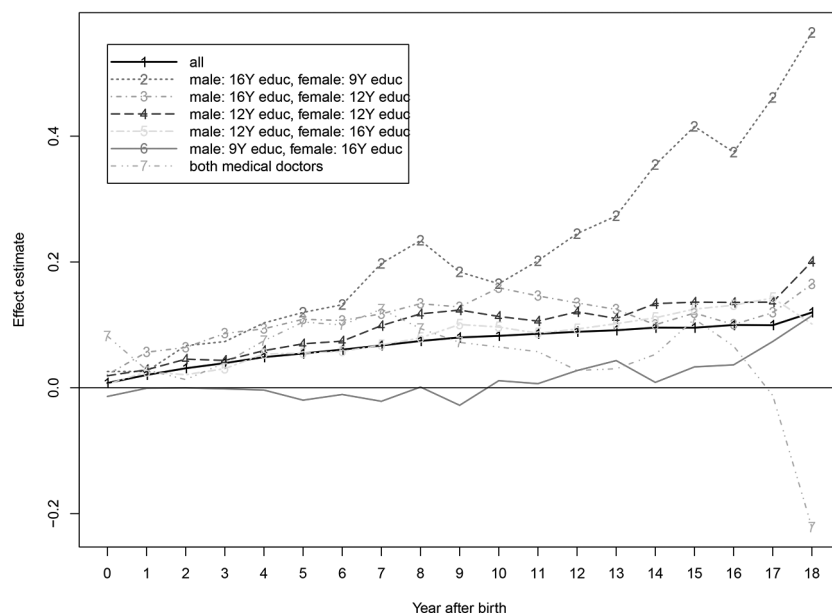


FIG. 6.—The effect of family formation on the change in the within-couple wage gap: effect size by relative within-couple education and separately for medical doctors. Years of education and completed MD degree are measured at pre-child levels at $t = -2$. Estimates from OLS regressions with $\ln y_{imt} - \ln y_{ift}$ for $t \geq -1$ as outcome variable, where $y_{imt/ift}$ is the yearly wage of the male/female within each couple in year t . A parameter estimate is significant at the 5% level if the corresponding group number is superimposed on the curve. For instance, the parameter estimate for group 7 is significant for the year of child birth but insignificant for year 10, and none of the estimates for group 6 are significant.

In the lower end of the conditional distribution, we have mothers with the most favorable counterfactual income and wage trajectories in relation to their partners. The opposite is true for couples in the upper part of the conditional distribution.

We start by presenting the results on the change in the within-couple gap in income over time. Figure 7 presents quantile regression estimates for the yearly effects for years 0, 2, 6, and 8.²⁴ The estimates from the quantile regressions for different percentiles are depicted together with confidence intervals (the shaded area). During the second year as parents, all couples experience a higher increase in the income of the males (year 2). The change in this gender gap is smaller the better the counterfactual income trajectory the mothers have in comparison to their male partners. In other words,

²⁴ In order to save space, we do not present results for intermediate years. The rest of the results follow the same pattern and are available upon request.

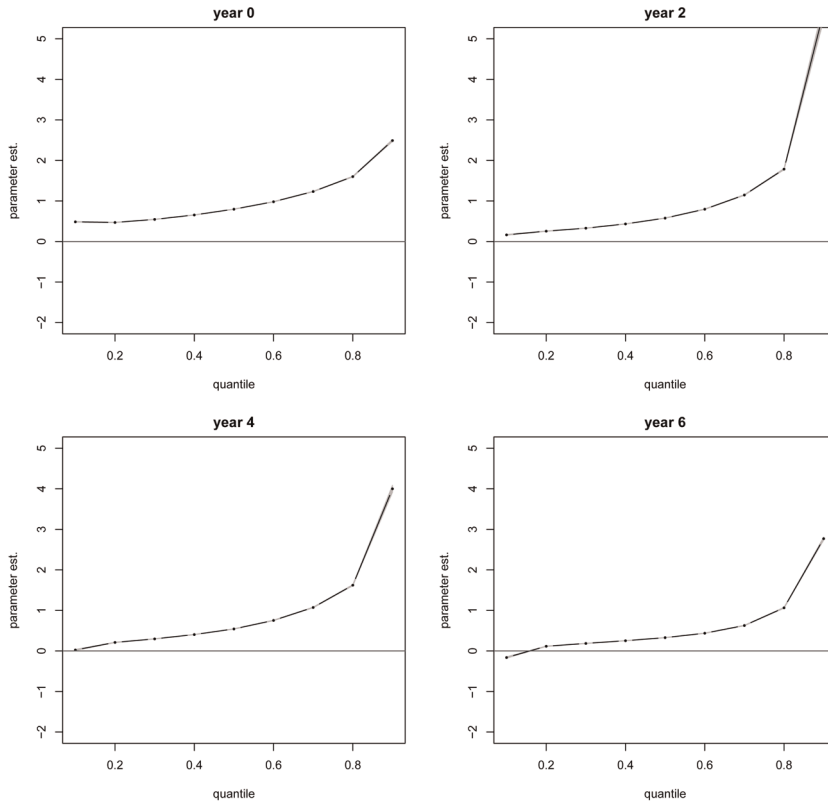


FIG. 7.—The effect of family formation on the change in the within-couple income gap: years 0, 2, 4, and 6. The effect is measured in various parts of the conditional distribution of the outcome variable $\ln y_{imt} - \ln y_{ift}$ for $t \geq -1$ with $y_{imt/ift}$ being the yearly income of the male/female within each couple in year t . The curve represents quantile regression estimates for different quantiles on the horizontal axis, with corresponding 95% CI (shaded area, barely seen due to high precision). The various graphs show parameter estimates of the yearly effect of having a child, measured as the percentage change in the gender gap in comparison to the pre-birth gap. The full set of controls consists of the pre-child gender log-difference in income, calendar year dummies, within-couple age gap, and pre-child within-couple differences in education, respectively.

mothers with higher (within-couple) relative earnings capacity bear smaller short-term economic costs from having a child, compared to mothers with lower relative earnings capacity. Looking at the longer run, we see that these heterogeneous effects are further enhanced. Starting from year 6, couples in the lowest part of the conditional distribution (the 10th percentile) even experience a decreasing gender gap over time as an effect of parenthood. In other words, in couples where fathers would have had the worst relative

income trajectory in the absence of children, fathers bear the largest economic cost of parenthood. In figure B1 in appendix B, we show the results for years 8, 11, 14, and 17, where we see a similar but more pronounced picture in the longer run. Here the gender gap decreases as a result of parenthood for roughly 20% of the couples.

The results from the quantile regressions on wages are presented in figure 8 (years 0, 2, 6 and 8) and in figure B2 in appendix B (years 8, 11, 14,

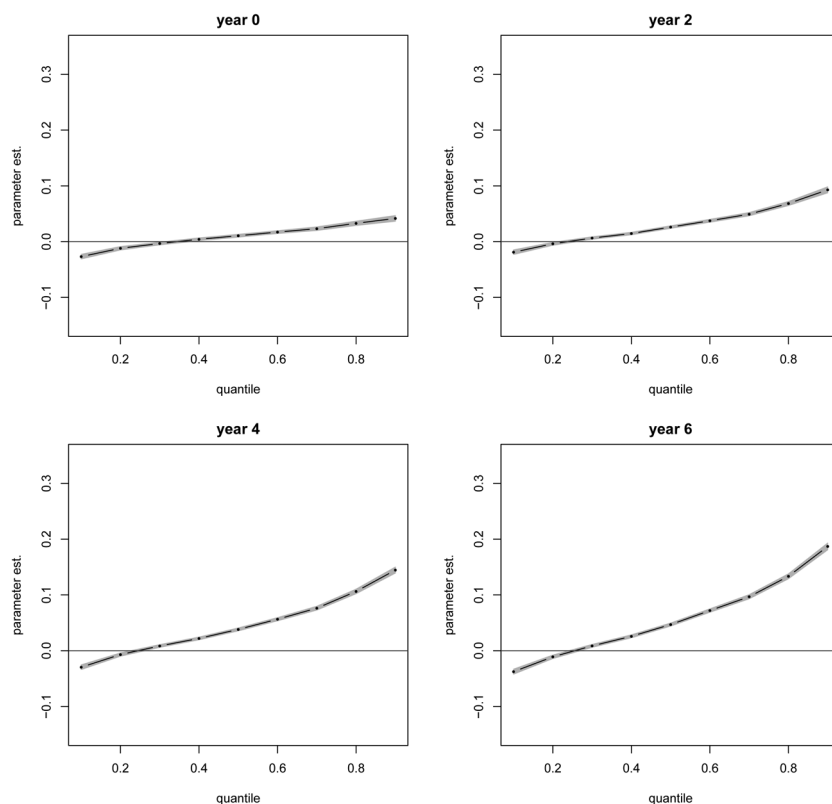


FIG. 8.—The effect of family formation on the change in the within-couple wage gap, years 0, 2, 4, and 6. The effect is measured in various parts of the conditional distribution of the outcome variable $\ln y_{imt} - \ln y_{ift}$ for $t \geq -1$ with y_{imt}/y_{ift} being the monthly wage of the male/female within each couple in year t . The curve represents quantile regression estimates for different quantiles on the horizontal axis, with corresponding 95% CI (shaded area, barely seen due to high precision). The various graphs show parameter estimates of the yearly effect of having a child, measured as the percentage change in the gender gap in comparison to the pre-birth gap. The full set of controls consists of the pre-child gender log-difference in wages, calendar year dummies, within-couple age gap, and pre-child within-couple differences in education, respectively.

and 17). We can draw the same conclusions from the results for wages as from the ones for income. The results are in line with the theoretical prediction on how the effect of parenthood varies across the conditional distribution if the decision of sharing the cost of parenthood depends on the comparative advantages within the couple. That is, the partner who has the smallest opportunity cost in terms of expected forgone earnings bears the largest cost of parenthood, here in terms of forgone wage increase.

This finding is in line with a collective labor supply model as in, for example, Blundell, Chiappori, and Meghir (2005). More recently, Cherchye, De Rock, and Vermeulen (2012) estimate a structural collective labor supply model with household production consisting of two domestic goods (parenting and general household work) on a rich time-use data set from the Netherlands. They are able to estimate the extent to which the husband's Pareto weight in the household bargaining increases with an increase in his relative wage within the household, and they find that the Pareto weight increases significantly with a relative wage increase. What our results show is that the relative potential earnings have a significant effect on whose labor supply decreases after the birth of the first child, and by how much.

VII. Discussion

We estimate both short- and long-term average effects of parenthood on the within-couple gender gap in income and wages. Furthermore, we focus on the importance of the matching of couples, that is, with whom one decides to form a family. We also estimate the effect separately for groups with varying relative within-couple education, and use quantile regressions. In this way, we study how the effect of parenthood differs across the distribution of the gender gaps in income and wages in the absence of having a child.

The main result is that parenthood affects men and women differently, which is in accordance with several earlier studies (e.g., Waldfogel 1997; Budig and England 2001; Sasser 2005; Kennerberg 2007; Loughren and Zissimopolus 2008). But in contrast to earlier studies, we ask a somewhat different question, namely: What happens to the within-couple gender gap in incomes and wages after the birth of the first child?

By restricting the question to the effects on the within-couple gender gap, we are able to control for both observed and unobserved attributes of the partner, which can be important if the labor supply of the partner changes as a response to parenthood. Using longitudinal data on the labor supply of women in a fixed-effect estimation of the effect of becoming a mother would potentially exaggerate the effect of parenthood on women's labor supply. The reason is that the change in the income of the partner is most likely a confounder in the model since the fathers' incomes theo-

retically affect the mothers' labor supply (Mincer 1966).²⁵ Our approach allows us to estimate the average effect of parenthood across the distribution of parents' potential earnings and to study the dynamics of the effect by means of evaluating the effect for every subsequent year after the birth of the first child.

We find that 15 years after entering parenthood, the percentage male-female gender gap in income has increased by 28 percentage points over its pre-child level. The corresponding increase for the gender gap in wages is 10 percentage points. The results estimated separately for groups with varying within-couple relative education, and from the quantile regressions, suggest that the effect of parenthood on the male-female gap increases with the income and wage gap the couple would have experienced in the absence of parenthood. The analyses confirm and extend on the results found in the case study on MBA students in Bertrand, Goldin, and Katz (2010). The general message is that the matchings of the couples are crucial for the magnitude of the effect of parenthood on the gender gap. That is, women's expected position within the family in terms of earnings potential is an important determinant for the size and sign of the effect of parenthood.

How do our main results compare with evidence from macrodata? We illustrate by using mid-career earnings for men and women. Taking data from the Swedish tax registers, we find that the percentage income difference between males of age 43 and females of age 41 was 47% in 2006.²⁶ We use 2006 for this example because in our data set 2006 is the year when we, on average in our sample, observe the incomes of mothers and fathers 15 years after child birth. Moreover, we show the percentage difference for the ages 43 for males and 41 for females because these are the average parental ages at which the first child turns 15 years. Using our data of matched couples, the percentage income difference between fathers and mothers 15 years after the birth of the first child is 53%. In a way, we are thus explaining more of the gender gap than can be found in macrodata. However, there is an obvious reason for this difference, namely, that we sample parents. As previously documented in Boschini et al. (2011, figure 12), the mid-career earnings of single men in Sweden are significantly lower than the corresponding earnings for fathers. Furthermore, although there is a similar difference between single women and mothers, it is much smaller. Thus, it should be expected that the mid-career gender gap for our sample of parents is greater than for the whole population including both parents and singles. Indeed, using official statistics once more, we find that

²⁵ Fathers' changes in labor supply can theoretically go in either direction: Kennerberg (2007), for example, finds that some fathers increase their labor supply when becoming a parent.

²⁶ The source is the table Louise for 2006, Statistics Sweden.

the percentage income difference between males of age 43 and females of age 41 was 62% for parents in 2006.²⁷ This number is larger than the 53% for parents in our data, but note that we have not taken into consideration the age of the child.

Loosely speaking, the observed gender income gap in the macrodata consists of a part that we explain in this paper, namely, the effect of parenthood, as well as parts due to a number of other factors. Not all individuals have children, and as mentioned above, the gender gap is much smaller for nonparents. Yet it is interesting to compare the magnitude of our parenthood effect estimate with the magnitude of the observed gender gap in the macro data. It follows from the discussion above that our baseline estimate of a 28 percentage points increase in the within-couple income gender gap explains about three-fifths of the 47% gender gap observed in the macro data.

Compared with previous studies of the gap between men and women within the same occupation, our average estimate is higher. Gender differences in family responsibilities are claimed to explain 39% of the gender gap among medical doctors (Sasser 2005) and 40% of the gender gap among lawyers (Wood, Corcoran, and Courant 1993). However, given our previous results, this difference is to be expected. As our quantile regressions show (see Sec. VI), mothers with lower relative earnings capacity bear higher economic costs from having a child than do mothers with higher relative earnings capacity. Arguably, in equally educated and high-skilled couples where both spouses are either physicians or lawyers, women's relative income trajectory in the absence of a child is higher than the average in the population. When we estimate our model separately for medical doctors, the effect of parenthood is in fact lower than for the whole population, although the point estimates after year 10 are insignificant due to the smallness of the sample.

We also find small short-term effects on the wage gender gap. Moreover, these effects increase with the age of the first child: after 15 years, women have 10% lower wages due to parenthood. Albrecht et al. (1999) and Datta Gupta and Smith (2002) study the effect of career interruptions due to child care, holding experience (labor supply) constant. These studies do not find any wage penalty of parental leave, at least not among women.²⁸ Since earlier studies indicate no wage penalty for women of parental leave, we interpret our negative estimates on the gender gap in

²⁷ The corresponding number for nonparents is 19%. The source is the table Louise for 2006, Statistics Sweden.

²⁸ Albrecht et al. (1999) study whether the effect of a career interruption differs depending on the reason for the break, and they conclude that there is a wage penalty of parental leave for men but not for women. Their explanation is that there are different signal values of being on parental leave for men than for women.

wages as an indirect effect of a gender differential effect of parenthood on effort in the labor market in terms of labor supply and labor market commitment. The Swedish social insurance allows for a generous job-protected leave: Swedish parents are allowed, with job protection, to reduce their working hours up to 25% until the child turns 8 years old. This opportunity is in particular used by Swedish mothers (see Sec. II). Thus, the wage effect we find is probably not explained by human capital depreciation while on leave but rather to a less rapid human capital accumulation associated with part-time work among mothers, which is likely attributed to mothers' long-term continuing responsibilities for child rearing.

VIII. Conclusion

The take-home message from this paper is that so long as family responsibilities are unequally shared, the gender gap is not likely to close and not even to narrow significantly. Swedish women participate in the labor market almost to the same extent as men, due to a series of political reforms (see Sec. II). These policies were successful in getting women to participate in the labor market, but to close the gender gap, women would also need to work the same amount of hours as men. As our results on the effect of parenthood on the income gender gap show, there is a large drop in working hours after the birth of the first child. This drop has dynamic implications in the form of a gradual relative depreciation of human capital, as seen from the results on the wage gap. This is true even in a highly flexible labor market such as the Swedish market, with its employment protection and legal rights to part-time work during the child's first 8 years.

Further, our results show the importance of comparative advantages within the household for the magnitude and sign of the effect of parenthood on the gender gap. We find that women whose husbands are relatively better paid or more educated take greater responsibilities at home. The reverse is also true, and our results reveal that in the long run, the gender gap decreases as a result of parenthood for approximately one-fifth of the households. We think that this is an important finding as it highlights the likely mechanism behind the observed effect of parenthood on the gender gap: the parent who is relatively better paid concentrates on labor market work, and the other parent picks up the slack at home. Since first-time fathers are on average 2 years older than their spouses, fathers should arguably have a comparative advantage on the labor market at the time of birth, all else being equal. However, there is a minority of households where parenthood actually decreases the gender gap because the females in those households have better opportunities on the labor market than their spouses have.

A recent explanation for the remaining gender gap is provided in Goldin (2014). Although our result of an increasing gender gap over the life cycle is in line with the empirical regularity discussed in Goldin (2014), we offer

a different explanation. Whereas Goldin (2014) argues that the most important measure for reducing the gender gap in pay is to increase workplace flexibility, we instead focus on the importance of the gender differential effects of parenthood, namely, that females take a larger responsibility for the household and the child-care duties. Workplace flexibility is a multi-dimensional concept. According to Goldin (2014), the term includes the number of hours to be worked and also the need to work particular hours, to be “on call,” provide “face time,” be around for clients, be present for group meetings, and so forth. Certainly technological advances that make it possible to work from home and be on call have facilitated combining family life and a career. However, if human capital evolution is an accumulative process that takes place at the workplace and women continue to take the largest responsibility for family duties, we claim that the gender differential wage trajectories over the life cycle will remain.

The two explanations for the remaining gender gap, namely, workplace flexibility and the effect of parenthood, have very different policy implications. Testing for their relative importance is therefore an endeavor that we plan to undertake in future research.

Appendix A

Empirical Strategy

Let $\{Y_{jt}(1)\}_{t=1}^T$, $j = m, f$ be the potential income (or wage) streams, expressed in logarithms, after becoming a parent for the males (m) and females (f), in time period $t = 0$. Further, let $\{Y_{jt}(0)\}_{t=1}^T$, $j = f, m$ be the corresponding income streams if not becoming a parent. Let $\{y_{jt}\}_{t=-1}^T$, $j = m, f$ be the observed income streams starting 1 year prior to parenthood. In the present paper, we match parents in $t = 0$ and we work directly with within-couple differences, defined as follows:

$$\begin{aligned}\tilde{Y}_t(0) &= Y_{mt}(0) - Y_{ft}(0), \\ \tilde{Y}_t(1) &= Y_{mt}(1) - Y_{ft}(1), \\ \tilde{y}_t &= y_{mt} - y_{ft}.\end{aligned}$$

Our estimand is

$$\alpha_t = E\{\tilde{Z}_t | \text{parent at } t = 0\}, \quad (2)$$

where $\tilde{Z}_t = \tilde{Y}_t(1) - \tilde{Y}_t(0)$.

The gender difference in potential outcomes after having a child is thus by definition equal to

$$\tilde{Y}_t(1) = \tilde{Y}_t(0) + \tilde{Z}_t, \quad t > 0. \quad (3)$$

Assume that the income difference in the absence of a child follows the autoregressive process

$$\tilde{Y}_t(0) = \delta + \rho \tilde{Y}_{t-1}(0) + \varepsilon_t, t \geq -1, \quad (4)$$

where $0 \leq \rho \leq 1$ and ε_t is a white noise process. Using recursive substitution, we get

$$\tilde{Y}_t(0) = \phi_0 + \theta_0 \tilde{Y}_{-1}(0) + \epsilon_t, \quad (5)$$

where $\phi_0 \equiv \delta \sum_{j=0}^t \rho^j$, $\theta_0 \equiv \rho^{t+1}$, and $\epsilon_t \equiv \sum_{j=0}^t \rho^j \varepsilon_{t-j}$. Substituting for $\tilde{Y}_t(1)$ and $\tilde{Y}_{-1}(0)$ with the observed outcomes $\tilde{y}_t$ and $\tilde{y}_{-1}$, and using (3) and (5), yields

$$\tilde{y}_t = \phi_0 + \theta_0 \tilde{y}_{-1} + \tilde{Z}_t + \epsilon_t \text{ for } t \geq 0. \quad (6)$$

For a sample of n parents, it is thus possible to estimate α_t from (2) using ordinary least squares (OLS) on the following model:

$$\tilde{y}_{tci} = \alpha + \psi_c + \theta_0 \tilde{y}_{(-1)ci} + \alpha_t + \mathbf{x}_{(-1)i}' \beta + u_{tci}, t \geq 0, \quad (7)$$

where the calendar year is indexed by $c = 1989, \dots, 2008$, $\tilde{y}_{tc}$ is the income or wage gender gap t years relative to childbirth and observed in calendar year c , i is a couple index, $\tilde{y}_{(-1)ci}$ is the male-female gap in income or wages prior to childbirth, and $\mathbf{x}_{(-1)i}$ are pre-child covariates that we discuss below. Note that since we have variation in calendar times of child birth (i.e., several cohorts of first-time parents), we are able control for secular trends or shocks, ψ_c .

The OLS estimator $\hat{\alpha}_t$ is consistent for α_t as long as u_{tci} is strictly exogenous. That is, conditional on any secular trend, within-couple pre-child gap, and pre-child covariates, the timing of childbirth cannot be based on expectations about future shocks to the gender gap trajectory in the absence of a child. To explain, consider first the error term in equation (6), which by assumption is based on a white noise process and is therefore purely random. However, when we go to the empirical specification in (7), we need to consider in which way potentially unobserved factors enter u_{tci} , thereby causing inconsistency. First, since all of the couples in our sample become parents at $t = 0$, unobserved factors correlated with the decision of having a child cannot cause an endogeneity problem in (7). However, as mentioned above, the timing of childbirth might be a confounder. Assume for the sake of an example that (i) parenthood does not affect the gender gap in earnings and (ii) parents choose to have their first child at a time when they know that the female is going to experience a substantial downward shift in earnings relative to the male. In this extreme setting, the parents' choice of $t = 0$ in relation to the calendar time given by

the index c is completely determined by their expectation about the future trajectory of the gender gap during the coming calendar years. This means that we would not be estimating the effect of parenthood but rather a change in the gender gap that would have occurred even in the absence of a child. Therefore, the identification assumption has to do with the relation between the two time indices in (7), namely, time since birth t and calendar time c . Stated simply, the calendar time at which the parents choose to set $t = 0$ (e.g., the year 1995) cannot be based on their expectations about how the gender gap would have evolved over the following years (e.g., 1996, 1997, . . .) in the absence of a child, conditional on the gap in the year 1994 and the pre-child covariates.

We now motivate our choice of which pre-child covariates to include in the regression. The model states implicitly that if two spouses have the same pre-child income, then there should not be any systematically diverging trend in the potential gap in the absence of a child. However, we need to also take into consideration the parents' individual income trajectories. In particular, there is a difference between women and men concerning their age when becoming parents, and women are generally more educated than men, implying that we measure the pre-child gap in different parts of the parents' individual trajectories. This means that having the same pre-child income is not completely informative about the future gap trajectory, since the parents' individual income profiles generally depend on both their age and years of education. Therefore, we need to control for the within-couple differences in those two variables, resulting in $\mathbf{x}_{i(-1)} = ((\text{Education}_{m(-1)} - \text{Education}_{f(-1)}), (\text{Age}_m - \text{Age}_f))_i$.

For the sake of clear exposition and simplicity, pre-child observations were indexed by $t = -1$, or “(-1),” in the various expressions above. In the empirical application in the main text, we used $t = -2$ instead. The reason for this has to do with the data we use. An assumption when conditioning on pre-child income or wage differences is that pre-child observations should be unaffected by having a child or giving birth. However, since we have yearly data on the outcome variables (income or monthly wages), most of the mothers of children born before October in year t are going to be pregnant for a fraction of year $t - 1$, which might affect the income or wages measured in year $t - 1$. Therefore, we condition on $\tilde{y}_{(-2)i}$ in the estimation.

Appendix B

Figures

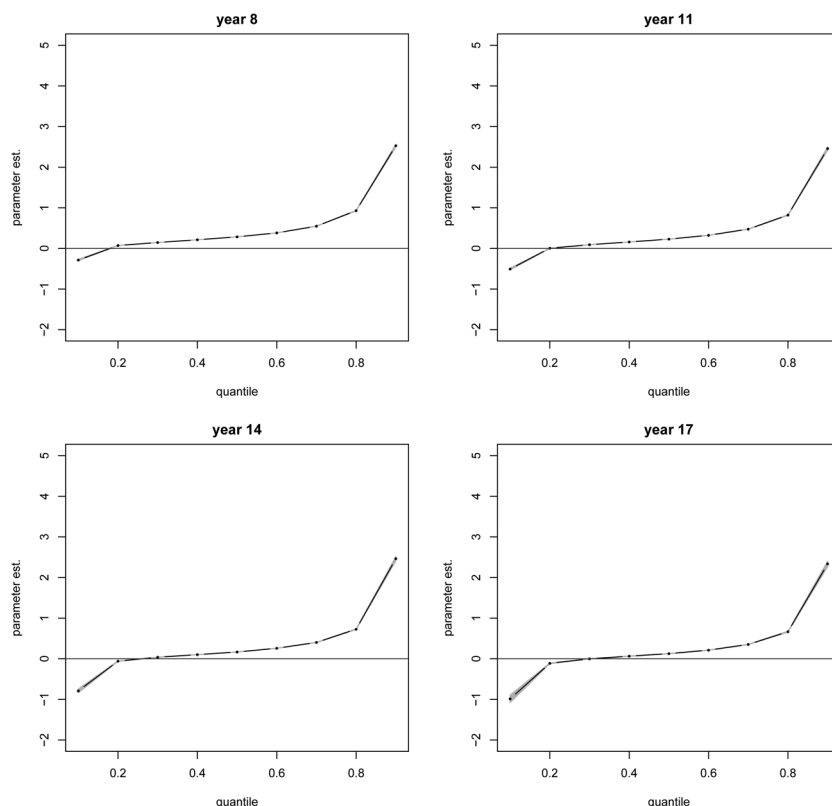


FIG. B1.—The effect of family formation on the change in the within-couple income gap: years 8, 11, 14, and 17. The effect is measured in various parts of the conditional distribution of the outcome variable $\ln y_{imt} - \ln y_{ift}$ for $t \geq -1$ with $y_{imt/ift}$ being the yearly income of the male/female within each couple in year t . The curve represents quantile-regression estimates for different quantiles on the horizontal axis, with corresponding 95% CI (shaded area, barely seen due to high precision). The various graphs show parameter estimates of the yearly effect of having a child, measured as the percentage change in the gender gap in comparison to the pre-birth gap. The full set of controls consists of the pre-child gender log-difference in income, calendar year dummies, within-couple age gap, and pre-child within-couple differences in education.

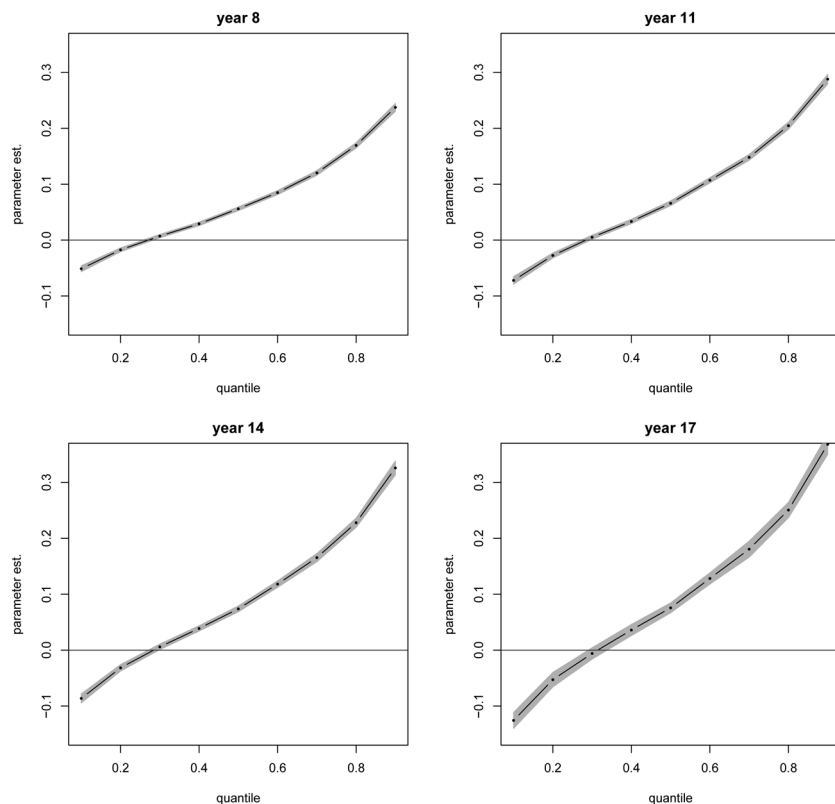


FIG. B2.—The effect of family formation on the change in the within-couple wage gap: years 8, 11, 14, and 17. The effect is measured in various parts of the conditional distribution of the outcome variable $\ln y_{imt} - \ln y_{ift}$ for $t \geq -1$ with $y_{imt/ift}$ being the monthly wage of the male/female within each couple in year t . The curve represents quantile-regression estimates for different quantiles on the horizontal axis, with corresponding 95% CI (shaded area). The various graphs show parameter estimates of the yearly effect of having a child, measured as the percentage change in the gender gap in comparison to the pre-birth gap. The full set of controls consists of the pre-child gender log-difference in wages, calendar year dummies, within-couple age gap, and pre-child within-couple differences in education.

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